DP Computer Science: A 2.2.4 Network Segmentation

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1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define **network segmentation** and explain its **purpose**
- Analyse the **benefits** of segmentation for **security, performance, and management**
- Identify **techniques** used to achieve segmentation (routers, switches, firewalls, VLANs, subnetting)
- Explain how **subnetting** and **VLANs** contribute to network segmentation
- Analyse **real-world scenarios** and recommend appropriate segmentation strategies

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Key Vocabulary

Term	Definition
Network segmentation	Dividing a network into smaller, more manageable segments
Subnetting	Dividing a network based on IP addresses into smaller subnetworks (subnets)
VLAN (Virtual Local Area Network)	Logically grouping devices regardless of their physical location
Broadcast domain	A network segment where broadcast traffic is forwarded
Broadcast storm	Excessive broadcast traffic that overwhelms a network
Firewall	A device that controls traffic flow between network segments
Router	A device that connects and routes traffic between different networks

Term	Definition
Switch	A device that connects devices within a network; can support VLANs
Access Control List (ACL)	A set of rules that controls which traffic is permitted or denied
Zero Trust	A security model where no device or user is trusted by default

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IB ATL Skills

Skill	Description
Thinking	Critical thinking, analysis, problem-solving
Communication	Explaining concepts and collaborating in discussions
Self-Management	Organisation, time management
Research	Gathering information and evaluating sources (optional extension)

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2. What is Network Segmentation?

Definition

> **Network segmentation divides a network into smaller, more manageable segments. It isolates devices, controls traffic flow, and enforces security policies.**

Why Segment a Network?

Think of a large school:

Problem	Solution (Segmentation)
Everyone can access everything	Staff, students, and guests have different access rights

Problem	Solution (Segmentation)
One virus infects all devices	Infected segment is isolated; infection is contained
Network slows down due to too many users	Traffic is divided; congestion is reduced
Troubleshooting is difficult	Problems are isolated to specific segments

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Visual Overview

Unsegmented Network:

```text
┌─────────────────────────────────────────────────────────────────┐
│                    UNSEGMENTED NETWORK                         │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────┐    ┌─────────┐    ┌─────────┐    ┌─────────┐   │
│    │ Device  │    │ Device  │    │ Device  │    │ Server  │   │
│    │  (PC)   │    │ (Laptop)│    │ (Phone) │    │  (Data) │   │
│    └─────────┘    └─────────┘    └─────────┘    └─────────┘   │
│         │              │              │              │         │
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│                        │              │                        │
│                     ┌──┴──────────────┴───┐                    │
│                     │       SWITCH        │                    │
│                     └─────────────────────┘                    │
│                                                                 │
│    ⚠️ Everyone can access everything                           │
│    ⚠️ One virus infects all devices                            │
│    ⚠️ Traffic congestion                                       │
│                                                                 │
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```

Segmented Network:

```text
┌─────────────────────────────────────────────────────────────────┐
│                     SEGMENTED NETWORK                          │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌──────────────────────┐    ┌─────────────────────────────┐ │
│    │    STUDENT VLAN      │    │         SERVER VLAN         │ │
```

```
│    │  10.0.1.0/24        │    │      10.0.3.0/24         │  │
│    │  ┌─────┐ ┌─────┐    │    │       ┌─────────┐         │  │
│    │  │ PC  │ │Laptop│   │    │       │ Server  │         │  │
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│                         │                                     │
│                    ┌─────────┐                                │
│                    │ Router/ │                                │
│                    │ Firewall│                                │
│                    └────┬────┘                                │
│                         │                                     │
│    ┌────────────────────┴──────────────────────────────┐     │
│    │                    GUEST VLAN                       │    │
│    │                  10.0.2.0/24                       │    │
│    │            ┌─────┐ ┌─────┐ ┌─────┐                │    │
│    │            │Phone│ │Tablet│ │Laptop│               │    │
│    │            └─────┘ └─────┘ └─────┘                │    │
│    └───────────────────────────────────────────────────┘    │
│                                                               │
│   ✅ Isolated traffic                                        │
│   ✅ Different access rights                                  │
│   ✅ Threats are contained                                    │
│                                                               │
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```

3. Benefits of Network Segmentation

Security Benefits

Benefit	Description
Isolate Sensitive Data	Financial systems and personal data are separated from general network traffic
Contain Threats	A compromised device cannot infect other segments
Control Access	Different segments have different access rights
Reduce Attack Surface	Fewer devices can see each other
Monitor Suspicious Activity	Easier to detect unusual traffic patterns between segments

Benefit	Description
Enforce Zero Trust	No device is trusted by default; every access request is verified

Example: A hospital separates patient records (secure segment) from guest Wi-Fi (unsecure segment). If a guest device is compromised, the patient data remains safe.

Performance Benefits

Benefit	Description
Reduce Congestion	Traffic is limited to specific segments
Improve Application Performance	Critical applications get dedicated bandwidth
Reduce Broadcast Traffic	Broadcast messages are contained within each segment
Prioritise Traffic	Important traffic (voice/video) can be prioritised
Optimise Resource Allocation	Resources are allocated where they are needed most

Example: A university separates student traffic (high volume) from research servers (critical). Research servers perform consistently even during peak student usage.

Management Benefits

Benefit	Description
Simplified Troubleshooting	Problems are isolated to specific segments
Easier Policy Enforcement	Policies are applied to segments, not individual devices
Scalability	New segments can be added as the network grows
Simplified Auditing	Easier to track access to sensitive systems
Change Management	Changes affect only one segment, not the entire network

4. Techniques for Network Segmentation

Technique 1: Routers

Aspect	Description
Function	Connects and routes traffic between different networks/subnets
How It Segments	Each router interface connects to a different network segment
Example	A router connects a home network to the internet; it separates internal traffic from external traffic
Layer	Network Layer (Layer 3)

Technique 2: Switches

Aspect	Description
Function	Connects devices within a network; can support VLANs
How It Segments	Layer 3 switches can route between VLANs; standard switches forward traffic based on MAC addresses
Example	A managed switch creates VLANs to separate departments in an office
Layer	Data Link Layer (Layer 2) or Network Layer (Layer 3 for managed switches)

Technique 3: Firewalls

Aspect	Description
Function	Controls traffic flow between network segments
How It Segments	Firewalls enforce security policies; they decide what traffic is permitted or denied between segments

Aspect	Description
Example	A firewall sits between the internet and the internal network; it blocks malicious traffic
Layer	Network Layer (Layer 3) to Application Layer (Layer 7)

Technique 4: VLANs (Virtual Local Area Networks)

Aspect	Description
Definition	Logically groups devices regardless of their physical location
How It Segments	VLANs separate broadcast domains; devices in different VLANs cannot communicate unless routed
Key Feature	Devices in the same VLAN can communicate as if on the same physical network, even if located in different buildings
Security Benefit	Isolates different user groups (e.g., students, staff, guests)
Performance Benefit	Reduces broadcast traffic

VLAN Diagram:

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┌─────────────────────────────────────────────────────────────────┐
│                          VLAN DIAGRAM                          │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │                      SWITCH                            │  │
│    │                                                         │  │
│    │    ┌──────────────┐  ┌──────────────┐  ┌────────────┐  │  │
│    │    │   VLAN 10    │  │   VLAN 20    │  │  VLAN 30   │  │  │
│    │    │   Students   │  │    Staff     │  │   Guests   │  │  │
│    │    ├──────────────┤  ├──────────────┤  ├────────────┤  │  │
│    │    │  ┌───┐ ┌───┐ │  │ ┌───┐ ┌───┐ │  │ ┌───┐      │  │  │
│    │    │  │PC1│ │PC2│ │  │ │PC3│ │PC4│ │  │ │PC5│      │  │  │
│    │    │  └───┘ └───┘ │  │ └───┘ └───┘ │  │ └───┘      │  │  │
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│    └─────────────────────────────────────────────────────────┘  │
│                              │                                  │
│                          ┌───┴───┐                              │
```

```
│                  │ Router │                    │
│                  └───┬────┘                    │
│                      │                           │
│              ┌───────┴────────┐                │
│              │   Internet/Firewall│             │
│              └──────────────────┘              │
│                                                  │
└──────────────────────────────────────────────────┘
```

Technique 5: Subnetting

Aspect	Description
Definition	Dividing a network based on IP addresses into smaller subnetworks (subnets)
How It Segments	Each subnet is a separate broadcast domain; routing is needed for communication between subnets
Key Feature	Uses a subnet mask to define the network and host portions of an IP address
Security Benefit	Subnets isolate traffic; devices on different subnets cannot communicate without routing
Performance Benefit	Reduces broadcast traffic; improves routing efficiency

Subnetting Example:

Subnet	Network Address	Host Range	Broadcast Address	Purpose
1	10.0.1.0/24	10.0.1.1 – 10.0.1.254	10.0.1.255	Students
2	10.0.2.0/24	10.0.2.1 – 10.0.2.254	10.0.2.255	Staff
3	10.0.3.0/24	10.0.3.1 – 10.0.3.254	10.0.3.255	Servers
4	10.0.4.0/24	10.0.4.1 – 10.0.4.254	10.0.4.255	Guests

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5. Comparison: Subnetting vs. VLANs

Feature	Subnetting	VLANs
Based On	IP addresses	Logical grouping
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Segmentation Type	Logical (IP-based)	Logical (group-based)
Broadcast Domain	Separate per subnet	Separate per VLAN
Location Independence	No (based on IP range)	Yes (devices can be anywhere)
Routing Required	Yes (between subnets)	Yes (between VLANs)
Best Used For	Larger networks; ISP networks	Organisations; campuses; flexible grouping
Example	Different departments have different IP ranges	Employees in the same department but different floors are in the same VLAN

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6. Scenario-Based Activity

Scenario 1: Small Office Network

Problem: A small office has a file server with sensitive data. All employees and guests share the same network. There are performance issues during peak hours.

Issue	Proposed Solution
Security Risk	File server accessible to all users
Performance Issue	All traffic shares the same bandwidth

Recommended Strategy:

- **VLANs:** Create separate VLANs for:
  - **Employees** (general access)
  - **Servers** (restricted access)
  - **Guests** (limited internet-only access)
- **Subnetting:** Assign different subnets to each VLAN to further control traffic

Modified Network:

```text
┌──────────────────────────────────────────────────────────────┐
│                    SEGMENTED OFFICE NETWORK                  │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│    ┌─────────────────┐  ┌─────────────────┐  ┌──────────────┐ │
│    │   VLAN 10       │ │   VLAN 20       │ │   VLAN 30     │ │
│    │   Employees     │ │   Servers       │ │   Guests      │ │
│    │  10.0.1.0/24  │  │  10.0.2.0/24  │  │  10.0.3.0/24  │ │
│    │ ┌───┐ ┌───┐  │  │ ┌───────────┐ │  │ ┌──┐  ┌──┐  │ │
│    │ │PC1│ │PC2│  │  │ │File Server│ │  │ │P1 │ │P2 │  │ │
│    │ └───┘ └───┘  │  │ └───────────┘ │  │ └──┘  └──┘  │ │
│    └──────────────┘  └───────────────┘  └─────────────┘  │
│                                                           │
│                ┌──────────────────────┐                   │
│                │  Router / Firewall   │                   │
│                │    (Access Control)  │                    │
│                └──────────────────────┘                   │
│                          │                                  │
│                     ┌────┴────┐                          │
│                     │ Internet│                          │
│                     └─────────┘                          │
└──────────────────────────────────────────────────────────────┘
```

Scenario 2: Home Network with IoT Devices

Problem: Smart home devices (cameras, thermostats, smart TVs) are on the same network as personal computers. IoT devices are often less secure and more vulnerable.

Issue	Proposed Solution
Security Risk	Compromised IoT device can access personal data
Unauthorised Access	IoT devices may have default passwords

Recommended Strategy:

- **VLANs:** Create separate VLANs for:
  - **General devices** (computers, smartphones)
  - **IoT devices** (smart TVs, thermostats, cameras)
- **Firewall Rules:** Block traffic from IoT VLAN to personal device VLAN

Modified Network:

```text
```

```
┌──────────────────────────────────────────────────────────────┐
│                   HOME NETWORK WITH IOT VLAN                  │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│    ┌─────────────────┐       ┌─────────────────┐           │
│    │   GENERAL VLAN  │       │    IOT VLAN     │           │
│    │   10.0.1.0/24  │      │  10.0.2.0/24  │           │
│    │ ┌───┐  ┌────┐ │      │ ┌─────┐  ┌─────┐ │           │
│    │ │PC │  │Laptop││      │ │Camera│  │TV   │ │           │
│    │ └───┘  └────┘ │      │ └─────┘  └─────┘ │           │
│    └───────────────┘      └─────────────────┘           │
│                                                          │
│                ┌─────────────────────┐                   │
│                │  Router / Firewall  │                   │
│                │  (Block IoT → LAN)  │                   │
│                └─────────────────────┘                   │
│                          │                                   │
│                     ┌────┴────┐                          │
│                     │ Internet│                          │
│                     └─────────┘                          │
└──────────────────────────────────────────────────────────────┘
```

Scenario 3: University Campus Network

Problem: A large university needs to manage thousands of students, faculty, staff, and guests. Different groups need different levels of access.

User Group	Needs
Students	Online learning platforms, library resources, internet
Faculty	Research databases, teaching resources, administrative systems
Staff	Administrative systems, email, work-related resources
Guests	Limited internet access only

Recommended Strategy:

- **VLANs:** Create separate VLANs for each user group
- **Subnetting:** Assign different subnets to each VLAN
- **Traffic Prioritisation:** Prioritise critical applications (research databases, online exams)
- **Access Controls:** Implement stricter controls for sensitive data (student records, financial info)

Network Design:

VLAN	Purpose	Subnet
VLAN 10	Students	10.0.10.0/24
VLAN 20	Faculty	10.0.20.0/24
VLAN 30	Staff	10.0.30.0/24
VLAN 40	Guests	10.0.40.0/24
VLAN 50	Servers	10.0.50.0/24

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7. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	10 min	"How can you ensure not everyone has access to every file?" Discuss chaos and security risks of unsegmented networks
Explore	20 min	Benefits of segmentation (security, performance, management); techniques (routers, switches, firewalls)
Explain	15 min	Subnetting and VLANs; how they contribute to segmentation
Interactive Activity	10 min	Scenario analysis—identify risks and propose segmentation strategies
Wrap-up	5 min	Key takeaways; exit ticket

Activity: Scenario Analysis

Instructions: For each scenario, identify potential security risks and performance bottlenecks, then propose a segmentation strategy.

1. **Small Office Network** — File server with sensitive data; employees and guests share the same network
2. **Home Network** — IoT devices on the same network as personal computers

3. **University Campus** — Thousands of students, faculty, staff, and guests with different access needs

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8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Scenario Analysis	Assess ability to identify risks and propose segmentation strategies
Quiz	Multiple-choice quiz on eBook page A2.2.4
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is network segmentation?
2. Why is segmentation important for security?
3. What is the difference between a subnet and a VLAN?
4. Name two techniques used for network segmentation.

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9. Differentiation

Level	Strategy
Support	Provide simplified diagrams, explanations, and partially completed diagrams
Challenge	Encourage research on real-world case studies or explore SDN and segmentation integration

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10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.2.4 with quiz)
- **Worksheet:** Scenario analysis + diagram design
- **Quizlet** (A2.2.4 vocabulary flashcards)

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11. Summary: Key Takeaways

Concept	Key Point
Segmentation	Divides a network into smaller, manageable segments
Security	Isolates sensitive data; contains threats; reduces attack surface
Performance	Reduces congestion; prioritises traffic; improves application speed
Management	Simplifies troubleshooting; easier policy enforcement
Subnetting	Divides network based on IP addresses (Layer 3)
VLANs	Logically groups devices regardless of physical location (Layer 2)
Firewalls	Control traffic flow between segments
Routers/Switches	Connect and separate network segments

> *"Network segmentation is essential for modern network design—it improves security, boosts performance, and simplifies management."*